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Comparison of methods for the processing of voltammetric electronic tongues data

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Abstract We are making a numerical comparison of various preprocessing strategies for dealing with data from voltammetric electronic tongues in order to reduce the high dimensionality of the response matrices. Different modelling tools are presented and briefly described. We then compare combinations of four preprocessing strategies (principal component analysis, fast Fourier transform, discrete wavelet transform, voltammogram-windowed slicing integral) with four modelling alternatives (principal component regression, partial least squares regression, multi-way partial least squares regression, artificial neural networks) by employing data from a voltammetric bioelectronic tongue, an array formed by enzyme-modified biosensors and applied to the discrimination and quantification of phenolic compounds.

Keywords Electronic tongue · Artificial neural network · Partial least squares regression · Multivariate data analysis · Data preprocessing · Voltammetric sensors

Introduction

In the recent years, there has been a demand of new analytical methods with the high sensitivity and selectivity, and fast response needed to meet new challenges in environmental monitoring, food safety and public health. In this fashion, industry is increasingly focused on fast-response and low-cost methods, as those based on chemical sensors, that may be used for screening or detecting any adulteration or contamination of the products, either during or after its production, or to assess they guarantee quality control standards [1, 2].

In this sense, classical research lines in the field of chemical sensors have focused on the development of ever more selective devices towards a particular species, and sensitive to lower concentrations at the same time. Unfortunately, there are few optimally operating chemical sensors that function without any interference or matrix effect in the required conditions.

The difficulty to obtain sensors with appropriate selectivity and sensitivity for a given analyte has led to the appearance of new working strategies. Accordingly, from the late 1990s a new concept in the field of sensors has appeared to tackle these problems: Electronic Tongues (ETs) [3]. These analytical systems are inspired by the sensory ability of taste in mammals, where a few receptors can respond to a large variety of substances. This principle is coupled with complex data treatment as applied in the brain, which allows to quantify or to classify a large amount of substances. These biomimetic systems, opposed to conventional approaches, are directed towards the combination of low selectivity sensors array response (or cross response features) in order to obtain some added value in the generation of analytical information.

According to the agreed IUPAC definition [4], an Electronic Tongue is "a multisensor system, which consists of a number of low-selective sensors and uses advanced mathematical procedures for signal processing based on Pattern Recognition (PARC) and/or Multivariate data analysis [artificial neural networks (ANNs), principal component analysis (PCA), etc.]". Hence, these systems consist of the coupling of an array of sensors capable of giving a wide and complete response of the analyzed species, plus a chemometric processing tool able to interpret and extract meaningful data from the complex readings, relating them with their analytical meaning [5]. Moreover, the data processing stage may offset any matrix or interference

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effect from the sample itself or allow the resolution of the interferences, drifts or non-linearity obtained with the sensors [6]. Thus, with this methodology, it is even possible to achieve a parallel determination of a large number of different species, while interference effects are solved by counterbalancing their presence [7].

Until now, ETs have provided alternatives to many classical methods, especially in the food analysis field [2, 5, 8]. Besides the wide variety of samples which they are applied to, there are two key parameters that determine the information obtained; i.e. the modelling tool and the type of sensors used.

On that account, data analysis and pattern recognition, are a fundamental part of any sensor array system [9]. This computer modelling stage is usually realized in two steps. First, the data is preprocessed in order to make it independent from units, remove redundant information and to enhance signal-to-noise ratio. Following this, the model describing the relation between readings and outputs is then created. Among the various numeric procedures that can be implemented, Principal Component Analysis (PCA), Partial Least Squares (PLS) and Artificial Neural Networks (ANNs) are the most widely used for ETs applications [2]. Concretely, PCA is the most common one being used either as a qualitative visualization tool or as a preprocessing step; whereas most advanced qualitative modelling may be achieved with the use of PLS-Discriminant Analysis (PLS-DA), Linear Discriminant Analysis (LDA) or Support Vector Machine (SVM). While in the case of quantitative applications, easier models can be built either using Multiple Linear Regression (MLR) or Principal Components Regression (PCR) [10], although better results can be achieved by using more powerful methods such as PLS or ANNs. Lastly, the new trends in data analysis are mainly related to the use of trilinear approaches such as PARAFAC or Tucker models for qualitative analysis, and Multi-Way PLS (*nPLS*) for quantitative models [11].

On the other hand, among the different families of chemosensors that may form the ET, potentiometric, voltammetric or impedimetric sensors are the main variants used [5, 12]. Furthermore, it is even possible to combine the responses obtained from different sensors families, a strategy known as data fusion [13]. Nevertheless, in this work we will focus on voltammetric Electronic Tongues, one of the most complex cases, given the high dimensionality of the considered signals which hinders its treatment, usually requiring the use of a preprocessing stage as stated below.

Therefore, by using ETs it is possible to provide multicomponent analytical information in real time with a direct, relatively simple measuring setup and low cost. In the same way, it is feasible to overcome the difficulties of analyzing the sensors' raw data by building calibration computer models. Some previous studies dealing with the preprocessing of data from voltammetric ETs have been reported, although they are mainly focused on classification tasks rather than quantification [14–17]. On that account, in this work we will focus on the different preprocessing strategies that can be used to reduce its dimensionality (i.e. compression of the original signal) and the different multivariate calibration methods (i.e. modelling sensors response) found in the literature. Finally, in order to illustrate the differences between different preprocessing options (e.g. PCA, FFT or DWT) and modelling methods (i.e. PCR, PLS, nPLS and ANN), a whole comparison of the performance of the proposed strategies was applied to the discrimination and quantification of phenolic compounds mixtures with data collected from a voltammetric biosensors array.

Experimental

Data preprocessing - compression

A known problem when voltammetric sensors are used is the large dimensionality of the generated data which hinders their treatment. That is, when a complete voltammogram is recorded for each sensor from the array. This is perhaps the main reason why this approach has been rarely used in the literature; especially if ANNs are to be used, in which case departure information needs to be preprocessed. Although signal preprocessing is not always strictly necessary (e.g. the whole data set may be employed in the case of PLS or PCR), it has been demonstrated that even in these cases, its use improves model predictive behaviour [18, 19]. Hence, this stage is necessary given the sensor array contains complex voltammetric responses encompassing global information about the compounds analyzed through the entire voltammetric curve. Despite differentiated response in terms of specificity being found with conventional amperometry treatment (currents measured at a selected potential), significant result improvements can be obtained by using the whole curve instead of particular peaks [20].

The straightforward solution is the use of multi-way processing methods (samples x sensors x polarization potential) like nPLS, but the intricacy of the approach is also critical [21, 22]. While still another option is the unfolding of original data into a 2D array [23], and the use of standard linear or non-linear chemometric calibration tools such as Principal Component Regression (PCR), Partial Least Squares regression (PLS) or Artificial Neural Network (ANN) models, this approach is normally avoided, given the distortions it introduces in the data.

In this sense, an attractive solution when dealing with a set of voltammograms is to employ a preprocessing stage for data reduction prior to modelling (Fig. 1). The main objective of this step is to reduce the complexity of the input signal preserving the relevant information and making it

compatible with ANNs or other numeric modellers, which facilitates an advantageous reduction in training time, to avoid redundancy in input data and to obtain a model with better generalization ability [24]. This compression stage may be achieved by the use of methods such as feature selection [25], Principal Component Analysis (PCA) [26], "kernels" [27], Discrete Wavelet Transform (DWT) [28] or even Fast Fourier Transform (FFT) [29].

Feature selection

The simplest solution in these cases is the mere selection of the data points that most contribute to the analyte signals and their inclusion into the modelling stage. Even though using the whole voltammetric curve always conveys useful information, there are advantages in identifying and removing the less-relevant inputs even if a large ANN or any other model is well trained: better modelling accuracy and higher robustness than the initial model; knowledge acquisition for the identification of the important inputs; larger ratio of examples to model connection weights (thus a better generalization capacity); ease of model comprehension [24]. Thus, better models are obtained with a reduced input set. For these reasons, different strategies for this feature selection, that attempt to capture the most important data points, are presented.

– *Knowledge-based*: in some cases through accumulated experience of the analyst, and specially when working with biosensors or some modified sensors which present very specific peaks, the potential in which voltammetric response appears has been previously identified. Thus, the main relevant information along the voltammetric curve is focused in these points, and by including them to the modelling stage, reduction of the data dimension can be achieved. Despite the favourable results obtained in this way, this procedure is not the best choice given that it requires prior knowledge of the system, selection of the inputs is very subjective, it can not be ensured that all the information contained on the raw data is included and the more relevant points cannot always be easily identified.
– *Statistical parameters*/*Stepwise inclusion methods*: in the same sense, selection of the relevant data points that must be included in the modelling stage can be achieved by selecting them according to some statistical parameters such as: correspondence analysis (evaluation of correlation between the data points and removal of the correlated ones), factor analysis extraction [30], genetic algorithms [31], etc.
– *Causal Index analysis* (*CI*): another possibility for feature extraction is the usage of ANNs as feature selection tools [32]. The use of this method for variable selection is the location of an optimal set of inputs that can successfully classify or predict the desired outputs. This is a feed-forward network that combines a back-propagation algorithm which is used to train the network according to a learning rule [33]. Therefore, to carry out the dimensionality reduction and feature selection, the following steps are followed: (1) First of all, the ANN architecture should be selected prior to perform modelling steps. For this, the number of neurons in the hidden layer and the transfer functions between the layers are selected in advance to represent the intrinsic dimensionality of the dataset. This represents a limit on the number of hidden neurons in the initial ANN. Based on the experience and literature, the suggested number of neurons in the hidden layer should be already very close to the optimal size before the network reduction procedures are applied, usually five neurons is a good number. Thus, most of the effort is directed at reducing the number of network inputs since the number of outputs is fixed by the requirements of the model. (2) Afterwards, it comes the determination of

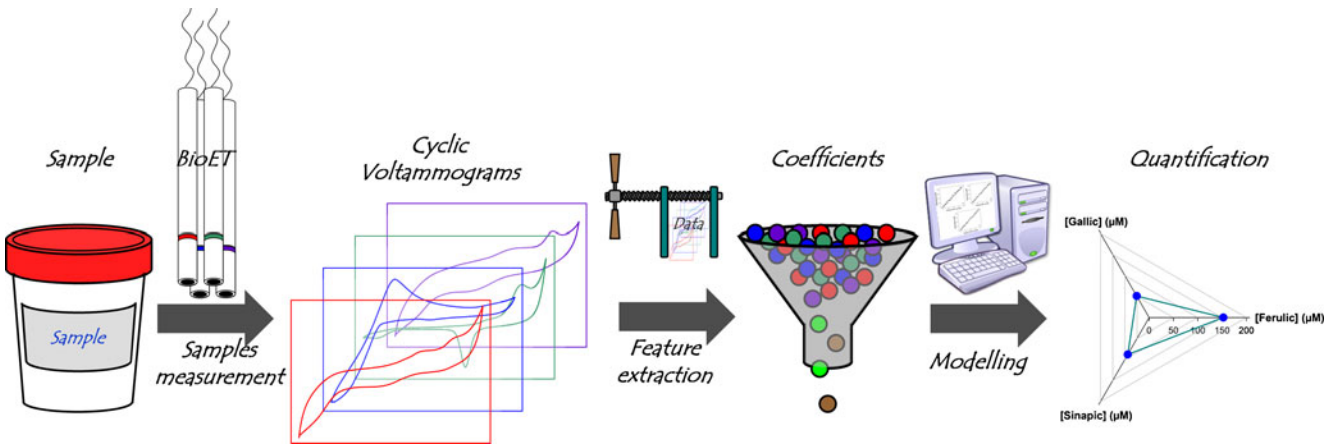


Fig. 1 Processing scheme for building the quantification model

the relevance of network inputs. Once the ANN model is trained, the analysis of its connection weights can easily identify the important inputs. In this manner, the relative contributions of the inputs to the variance of the output layer in the trained network are calculated [24]. Inputs that make relatively small contributions to the variance can be discarded, viz. pruned. These small contribution values indicate that either the input does not change significantly or that the training of the ANN has allocated small connection weights to scale the input. (3) Repeating the training process of the ANN model with the reduced input set and selecting the most relevant inputs can proceed until a near-optimal, small, set of inputs is identified [24, 32]. Consequently, better models are built with the reduced input set, and this process is repeated until the model accuracy is degraded. A known drawback is that some of the resulting minimal sets are not unique, which will depend on the ANN architecture used in the training.

Voltammogram windowed slicing integral

This method considers the area under the voltammetric response curve, which can also be interpreted as the scaled average of the data points [27]. Thus, in this case the voltammogram is divided into k sections and the area under each section is used as input data point to the modelling stage.

The idea behind this compression technique is to capture some information about the voltammetric waveform characteristics. The number of sections (k) into which voltammogram is divided is arbitrary and it depends on the response profile of the data; by increasing k, the original information in the voltammetric data is best reproduced, but at expenses of the compression ratio. When the k value is close to the number of points in the original signal, the obtained coefficients are just a scaled value of the original signal; that is, the original value multiplied by a constant factor. It is for this reason that a compromise between the compression ratio and signal reproduction must be taken into account.

Principal Component Analysis (PCA)

A viable option for data reduction is the use of PCA as feature extraction tool. This is a useful procedure able to reduce the dimensionality of large data sets, such as those from voltammetric sensor arrays; and it might also be used to find signals in noisy data.

PCA allows for the projection of the information carried by the original variables onto a smaller number of underlying ("latent") variables called principal components (PCs) with new coordinates called scores, obtained after data transformation. The first PC covers as much of the variance in the data as possible; the second principal component is orthogonal to the first and covers as much of the remaining variation as possible, and so on. Then by plotting the PCs, one can view interrelationships between different variables, and thus detect and interpret sample patterns, groupings, similarities or differences [34].

As in the previous cases, there is a compromise in the number of extracted features (PCs in this case) used as inputs into the modelling stage. In this case, the accumulated variance could be used as a measure of the reconstruction degree, obtaining an indicator of the number of PCs that must be taken into account.

Fast Fourier Transform (FFT)

FFT is an efficient algorithm to compute the Discrete Fourier Transform (DFT) much faster; the reason might be explained by the fact that computing it directly from the definition is often too slow to be practical, thus with the use of FFT, the difference in speed can be substantial, especially for long data sets (in practice, the computation time can be reduced by several orders of magnitude) [35]. DFT is an important tool in digital signal processing which decomposes a sequence of values into components of different frequencies.

The processing of the signals using Fourier transform could be considered as the decomposition of the signal using a sine/cosine function pair at different frequencies, calculating a coefficient for each one taking into account its contribution to the original signal. When DFT analysis is applied for signal decomposition, an equal number of coefficients and points from the original signal are obtained, where every contribution represents the information of frequency bands contained in a specific segment of the analysed signals. As in the case of PCA, the most relevant information is kept by the first coefficients, ergo good signal reconstruction could be achieved just preserving the first coefficients and discarding the rest. In this treatment, the Nyquist theorem must be recalled, limiting the highest frequency that carries significant information to the discretization interval [36]. Moreover, an additional advantage of this approach is the facility to remove the noise present in signals thorough the removal of the high frequency contributions, normally related to it [37]. Finally, the number of coefficients to be kept can be optimized through the inverse Fourier transform, which can be applied to evaluate the degree of signal reconstruction.

Discrete Wavelet Transform (DWT)

The Wavelet transform (WT) is a high performance signal processing technique developed from the Fourier Transform during the late 1980s [38]; instead of projecting the signal

down to a cosine or sinus wave, the WT projects the signal down to a wavelet function, with the key advantage over FFT being its temporal resolution: it captures both frequency and location information (location in time). In the last decade, numerous applications of the WT have been proposed for chemical analysis, including denoising and smoothing, data compression, baseline correction, resolution of overlapping signals, electroanalytical chemistry and chromatography [39].

WT is used for signal decomposition onto a set of basis functions. These functions are obtained by dilations and translations of a unique function called mother wavelet [40], *Daubechies* wavelets being the most commonly used set of discrete wavelet transforms [37]. Thus, WT decomposes the signal in labelled and located contributions for one scale and one position parameter. This is implemented using Mallat's pyramidal algorithm [38], which operates over a single discrete signal of length M by decomposing it into orthogonal subspaces of length ca. $M/2$ in each step [28]. In this way, increasing the decomposition level (which means repeating the decomposition process) k times, the signal compression ratio can be increased. Nevertheless, the compression ratio is at enhanced at the expense of signal reconstruction, given signal reconstruction accuracy is degraded for each step.

Also, as in the case of Fourier transform, DWT presents the additional advantage of its ability to easily decrease the noise present in signals. Furthermore, WT can be used for data compression if only the most significant decomposition coefficients are kept for further treatment. WT has shown high efficiency in this situation, since it offers the advantage of performing data compression, feature extraction and noise reduction at the same time [28]. As in the previous case, this transformation can be undone to evaluate the signal reconstruction degree, which permits to optimize the number of inputs.

Data processing – modelling

After applying the desired preprocessing method, the next step is the modelling of the target variables using the obtained coefficients as inputs into the modelling stage. This modelling stage may be achieved through the use of linear methods such as MLR [18], PCR [41], PLS [21] or nPLS [21] or with non-linear ones such as ANNs amongst others [7]. This sorting takes into account the form in which regression analysis is done; that is, in the case of non-linear models, data is modelled by a function which is a nonlinear combination of the model parameters and depends on one or more independent variables, fitted by a method of successive approximations. While in the case of linear models, data is fitted using linear regression, usually by means of least squares approach. In the next sections, basic principles and discussion of the idea behind each modelling method are introduced.

Principal Component Regression (PCR)

PCR is a method for relating the variance in a target variable (y; usually the expected concentrations) to the variance of several predictors (x; response matrix of the ET array), for explanatory or predictive purposes. PCR is a two-step procedure which first decomposes the data response matrix with the use of PCA, and then fits a Multiple Linear Regression (MLR) model, using the PCs scores instead of the original response variables as predictors [34].

MLR is a classical regression method that combines a set of several predictors or response variables, in linear combinations, which correlate as closely as possible to a corresponding single response [10]. Thus, even the results for several predictors can be obtained in one run, these are based on individually built separate models. The same limitation is obtained in the case of PLS-1, where only 1 response variable could be modelled at once. In this sense, PCR could be considered as an alternative to PLS-1 [10].

This method performs particularly well when the various response variables express common information, i.e. when there is a large amount of correlation, or even collinearity [42]. Since the scores are orthogonal, the MLR solution is stable and therefore the PCR model does not suffer from collinearity effects. For its optimization, only the number of PCs used for the MLR regression must be taken into account, thus selecting those that lead to the lower RMSE achieved by the model for predicted values.

Partial Least Squares regression (PLS-2)

PLS is a very similar linear transformation procedure to PCR; it models both the ET response (x) and target (y) matrices simultaneously to find the latent (or hidden) variables in x that will best predict the latent variables in y. These PLS components are similar to PC's, but will be referred to as latent variables or factors [43]. Indeed, instead of finding hyperplanes of maximum variance between the response and independent variables, it finds a linear regression model by projecting the predicted variables and the observable variables to a new space. Because both the x and y data are projected to new spaces, the PLS family of methods are known as bilinear factor models.

The number appended at the end of the acronym identifies the number of modes of the responses, which in the case of PLS it is restricted up to 2 [23]. That is, PLS1 where one block of data consists of a single variable or PLS2, being both blocks multidimensional arrays. Accordingly, unlike PCR or PLS-1, PLS-2 analysis could be carried out with one or more y variables, meaning that multiple y responses can be modelled at once during regression stage.

PLS simultaneously decomposes x and y into a set of latent variables that maximizes the covariance between the

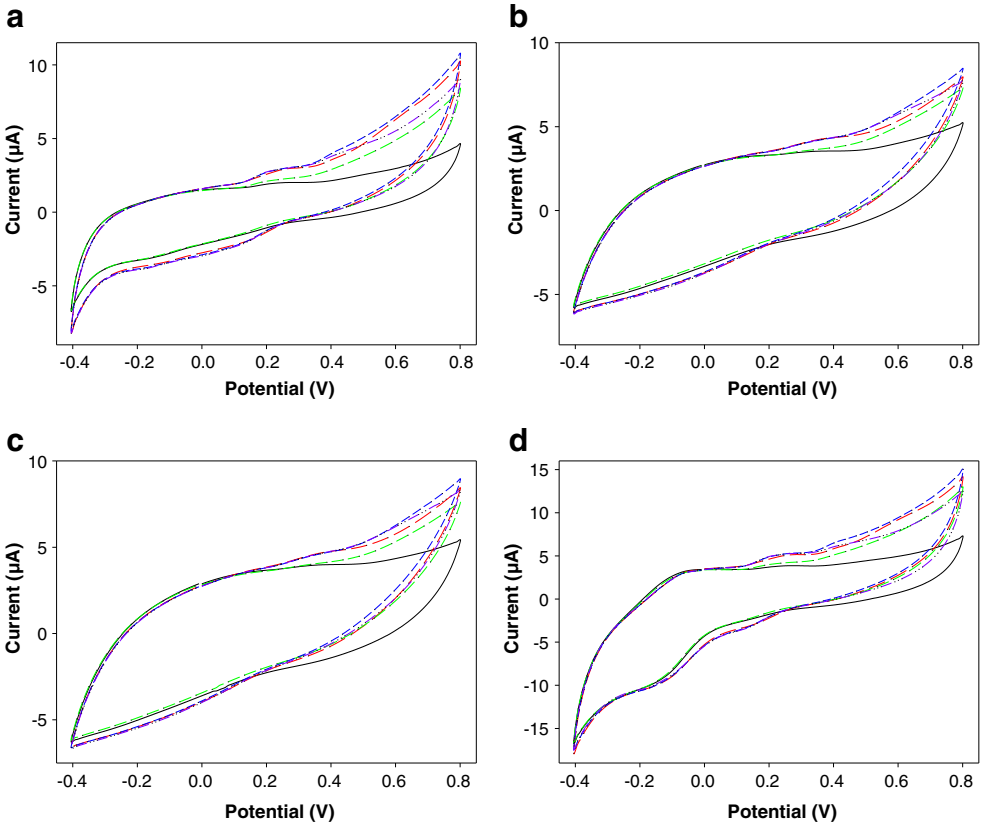


Fig. 2 Raw transient response of the 4-voltammetric sensors array: **a** graphite–epoxy sensor, **b** tyrosinase biosensor, **c** laccase biosensor and **d** copper nanoparticle modified sensor. Singals obtained for different mixtures of the three phenols are shown

independent and dependent data. In this case, convergence of the system to a minimum residual error is often achieved in fewer factors than using PCR. As before, only the number of latent variables must be optimized in order to get the lowest RMSE.

Multi-way Partial Least Squares Regression (nPLS)

This method is an extension of standard PLS regression methods and can be used to regress on data structures of a varying number of dimensions (multiway data or tensor) [11, 22]. As previously stated, this method has the advantage that it can relate the variations in one of several response variables (y) arranged in a 2D table to the variations of several predictors arranged in a 3D table (primary and secondary x variables), with explanatory or predictive purposes. Thus, it overcomes the disadvantages caused by the unfolding process, that is, the discontinuities created from one recording to the next, which causes the loss of the original relationship and creates limitations for first-order calibration [44].

As an extension to two-way PLS regression, n-PLS regression builds a calibration model incorporating a relationship between the sets of predictors and responses based on the multiway structure of the arrays (tensor). The main difference between n-PLS compared to two-way PLS regression is the use of loading matrices of higher dimensions for decomposing the multiway array (tensor). The goal of the methodology is to simultaneously fit multilinear models formed by score and loading vectors from both predictors and response arrays, along with a regression model relating the two decomposition models. According to the underlying theory of PLS regression, the model is fitted such that the score vectors have maximum covariance with the unexplained part of the dependent variable [23].

Table 1 Number of features used as input in the modelling stage and compression ratio for the different preprocessing techniques

	Number of features	Number of inputs	Compression ratio	Explained variance	R^2	*fc*
CI-ANN	–	25	97.7 %	–	–	–
Integral	11	44	95.9 %	–	–	–
PCA	4	16	98.5 %	99.9 %	–	–
FFT	16	64	94.0 %	–	0.999	0.975
DWT	23	92	91.4 %	–	0.997	0.969

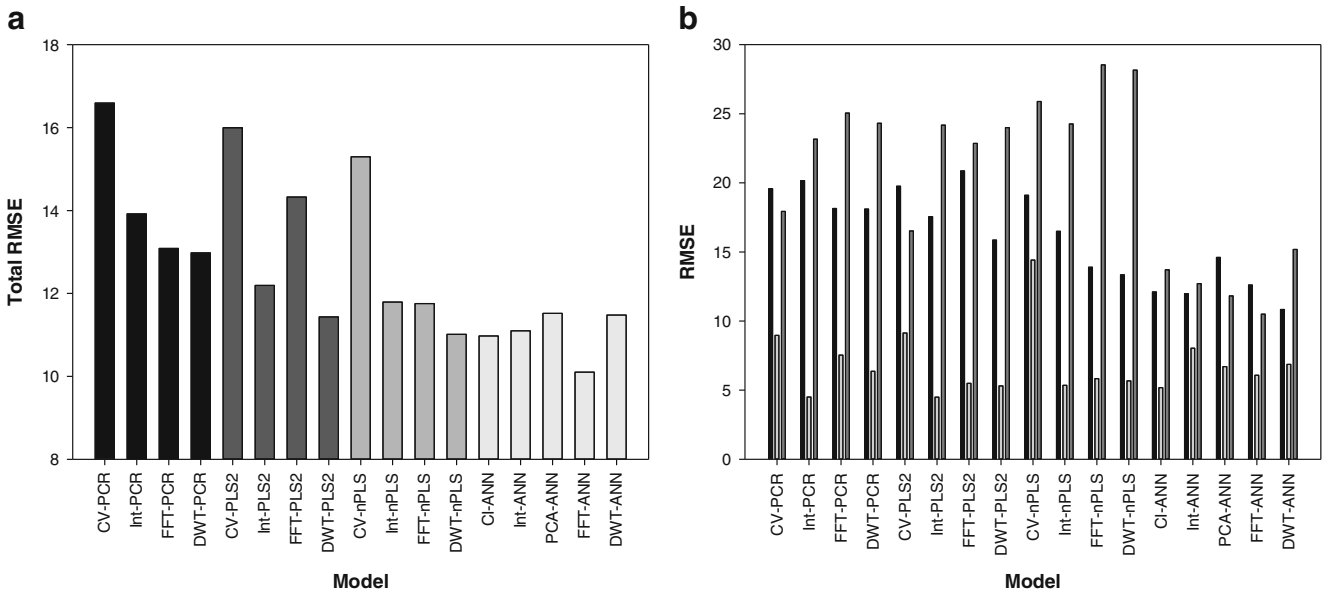


Fig. 3 Obtained RMSE values (expressed in μM) for the testing subset and the different models constructed. **a** Total RMSE and **b** individual RMSE for each of the considered phenols: (■) ferulic, (■) gallic and (■) sinapic acids

In addition to this, models obtained by applying first-order methods to data whose multiway structure is ignored can be less robust and hard to interpret, on the opposite to models based on multiway methods, which also yields better predictions and possibilities for exploring the data [22, 45].

Artificial Neural Networks (ANNs)

ANNs were originally designed to mimic the function of the human brain and applied to quantitative and qualitative analysis during the last decades [46]. They consist of a number of simple processing units (or neurons) linked by weighted modifiable interconnections [47].

Compared with previous presented methods, ANNs are more flexible modelling methodologies, since both linear and non-linear functions can be used (or combined) in the processing units, thus they are specially suited to being used with non-linear sensor responses. This also allows for more complex relationships between a high-dimensional descriptor space and the given retention data, and may lead to better predictive power of the resulting ANN model compared with other linear methods.

However, the major disadvantage of ANNs is also directly related to the complex model infrastructure, which requires some previous knowledge for the selection of the topology of the network and significant effort must be reckoned on the optimization due to the difficulties in predicting the optimum ANN configuration in advance [48]. This is a trial-error process where several parameters (training algorithms, number of hidden layers, transfer functions, etc.) are fine-tuned in order to find the best configuration that optimizes the performance of the neural network model. Moreover, as previously discussed, before building the ANN model a preprocessing step for reducing the input dimensionality is usually required.

Software

Chemometric processing was done by specific routines in MATLAB 7.10 (MathWorks, Natick, MA) written by the authors, using Neural Network Toolbox (v6.0.4), Wavelet Toolbox (v4.5) and Statistics Toolbox (v7.3). Sigmaplot 2000 (Systat Software Inc, California, USA) was used for graphic representations of data and results.

Results and discussion

In order to illustrate the differences between the various preprocessing and modelling methods, their performance was evaluated in the discrimination and quantification of polyphenol mixtures with data collected from a voltammetric BioElectronic Tongue employing an array of enzymatic biosensors [25] (details on sensors preparation can be found in Supporting Information).

The study case was the resolution of three phenols mixtures formed by ferulic, gallic and sinapic acids, three antioxidant compounds usually found in beers [49]. To this end, a total set of 42 samples were manually prepared with a concentration range for the three species from 0 to 200 μM for each polyphenol. The set of samples was divided into two data subsets: a training subset formed by 27 samples (64 %), which were distributed in a cubic design and used to establish the response model [25]; plus 15 additional samples (36 %) for the testing subset, distributed randomly along the experimental domain,

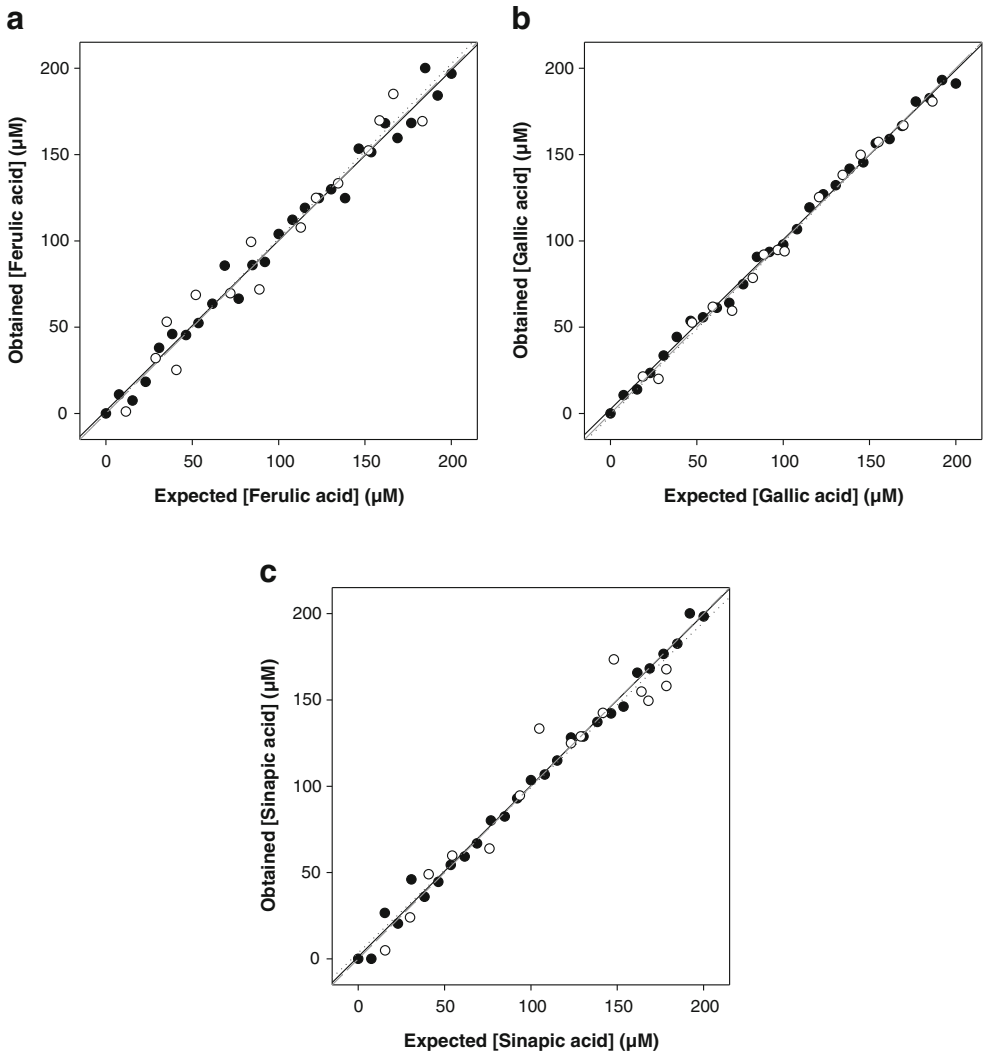


Fig. 4 Modelling ability of the optimized CI-ANN model. Comparison graph of expected vs. obtained concentrations for **a** ferulic, **b** gallic and **c** sinapic acids, both for training (●, *solid line*) and testing subsets (○, *dotted line*). Dashed line corresponds to theoretical diagonal line

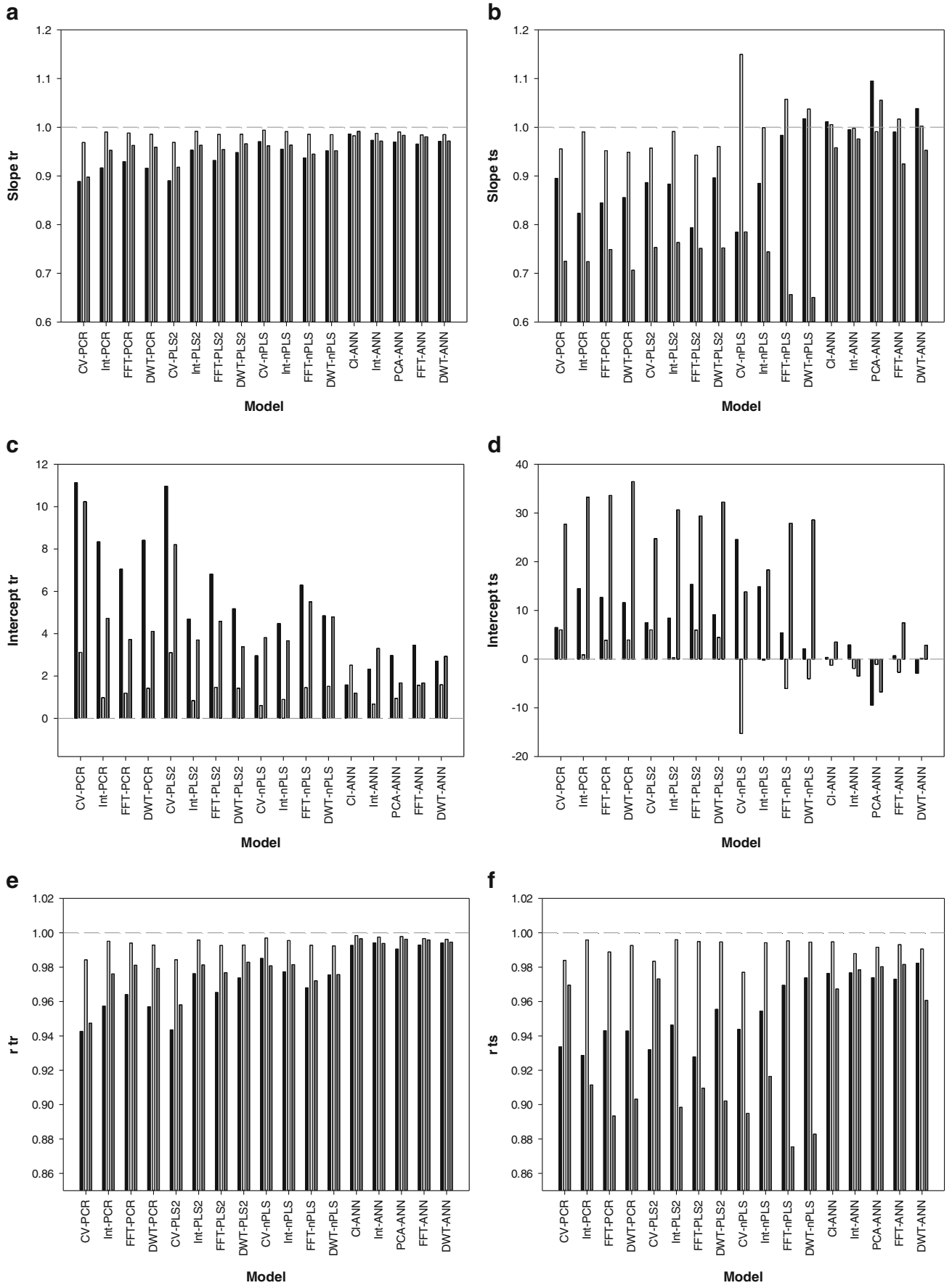
and used to evaluate the models predictive ability. In this way, the set of samples was measured employing the 4-sensor array, recording a cyclic voltammogram for each sensor. Examples of the different obtained signals for some of the phenolic mixtures are shown in Fig. 2. As can be seen, differentiated signals were observed for the different sensors, with complex and overlapping voltammetric responses that contain global information about the compounds analyzed along all the voltammogram. However the amount of information and the dimensionality of the signals (4 sensors x 268 current values x 42 samples) generate great difficulties in data treatment. Therefore, to fully exploit all the information obtained from each voltammogram, a compression step is required.

Data preprocessing - compression

In order to quantify the reconstruction degree in the cases where the signals could be reconstructed from the obtained coefficients, the coefficient of determination (R^2) and a comparison factor named *fc*, that considers the area under both signals when superimposed, where used [28]. The *fc* is defined as the ratio of the area intersected by both curves to the total area under both curves. From the set theory, with A and B as the areas under each curve, *fc* can be expressed as $fc = (A \cap B)/(A \cup B)$. This factor ranges from 0 to 1 depending on signals similarity; it values 0 when two signals have nothing in common and increases its value proportionally with similarity. In this sense, *fc* computes similarity in a way related to a correlation coefficient R, but being more sensitive to small differences.

Table 1 summarizes the results obtained after applying the different preprocessing methods. In the different cases, parameters of the algorithms were selected to ensure that practically all the information contained in the original data

Fig. 5 Fitted regression lines of the comparison between obtained vs. ▶ expected results attained with the different models for the three considered phenols ((■) ferulic, (■) gallic and (■) sinapic acids), both for training and testing subsets respectively. **a**, **b** Slopes, **c**, **d** Intercepts (μM) and **e**, **f** correlation coefficients


a
Slope tr
1.2
1.1
1.0
0.9
0.8
0.7
0.6
CV-PCR
Int-PCR
FFT-PCR
DWT-PCR
CV-PLS2
Int-PLS2
FFT-PLS2
DWT-PLS2
CV-nPLS
Int-nPLS
FFT-nPLS
DWT-nPLS
CI-ANN
Int-ANN
PCA-ANN
FFT-ANN
DWT-ANN
Model
b
Slope ts
1.2
1.1
1.0
0.9
0.8
0.7
0.6
CV-PCR
Int-PCR
FFT-PCR
DWT-PCR
CV-PLS2
Int-PLS2
FFT-PLS2
DWT-PLS2
CV-nPLS
Int-nPLS
FFT-nPLS
DWT-nPLS
CI-ANN
Int-ANN
PCA-ANN
FFT-ANN
DWT-ANN
Model
c
Intercept tr
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8
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2
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CV-PCR
Int-PCR
FFT-PCR
DWT-PCR
CV-PLS2
Int-PLS2
FFT-PLS2
DWT-PLS2
CV-nPLS
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PCA-ANN
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CV-PLS2
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DWT-PLS2
CV-nPLS
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Model
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1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
CV-PCR
Int-PCR
FFT-PCR
DWT-PCR
CV-PLS2
Int-PLS2
FFT-PLS2
DWT-PLS2
CV-nPLS
Int-nPLS
FFT-nPLS
DWT-nPLS
CI-ANN
Int-ANN
PCA-ANN
FFT-ANN
DWT-ANN
Model
f
r ts
1.02
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
CV-PCR
Int-PCR
FFT-PCR
DWT-PCR
CV-PLS2
Int-PLS2
FFT-PLS2
DWT-PLS2
CV-nPLS
Int-nPLS
FFT-nPLS
DWT-nPLS
CI-ANN
Int-ANN
PCA-ANN
FFT-ANN
DWT-ANN
Model

is preserved, as demonstrated in the case of PCA, FFT and DWT. Moreover, it should be noted that in all cases compression ratio was above 90 %, such a huge value will allow a reduction in training time and model complexity, as well as obtaining models with better generalization ability.

Beyond the compression ratio provided by each method, some of the advantages and disadvantages of each method should be taken into account and thus, depending on the case select the most appropriated one. For example, the use of CI is quite a time-consuming procedure, nevertheless it is a useful way to identify the significant features on the response profile; namely, which sensors provide more significant information and which potentials are associated with the electrochemical processes taking place. In the case of windowed slicing integral, the number of sections is arbitrary, thus in our case a value of k=11 was selected, despite good results were obtained even with less coefficients. This value was preferred given it was more comparable to the other methods, where a similar number of coefficients was used. In this sense and depending on the case, this parameter can be changed in order to improve the detail representation of the original signal or to reduce the number of inputs. The use of PCA is another simple way to reduce the high-dimensionality of the input signal, despite this method not being recommended in cases where there is some drift on the measurements given it is very sensitive to it, thereby providing coefficients more related to the drift than to samples information [50]. Also, its usage is not recommended in all the cases, given it focuses on the maximum variance between samples, hence giving less importance to small differences which in some cases could be significant. That is, if in our set we have a sample that is very different from the rest, that variance would not allow us to see the smaller differences between the rest of the samples (in this case, can't see the trees for the forest). Finally, even though it is not so clear in this case, it must be acknowledged that in complex and highly overlapped responses, some methods like FFT or DWT are more efficient [51]. The reason is due to the fact that the obtained coefficients are a representation of the entire voltammetric signal, which is also denoised during the feature extraction process.

On the other hand, it must be considered that the good results (large explained variance with only a few PCs) obtained with the PCA analysis are mainly due to the type of sensors used and the study case. That is, in this application the voltammetric ET was formed by an array which incorporated some specific biosensors selective towards phenolic compounds, and in addition synthetic samples were used. Thus, sensors' response is clearly related to desired compounds concentration and could be more easily identified at specific regions along the voltammogram. In this situation, some research groups [52] claim that the use of only one half of the voltammetric wave is enough given that following a reversible electrochemical process, the same information could be obtained either from the cathodic or the anodic waves. In this manner, the described preprocessing methods could be further simplified by just focusing on one part of the voltammetric response, either the cathodic or anodic component, or even a more restricted region.

Data processing - modelling

Applying the different preprocessing strategies to the raw data obtained from the ET sensors array, the different coefficients were obtained, which were used as inputs in the modelling stage. In the case of linear methods (PCR, PLS and nPLS), models were also built using the whole voltammetric curve as input; this was not done in the case of the ANN given this method is more sensitive to the number of inputs, consequently obtaining less robust models [48]. In this case, CI analysis was applied, which is based on the use of the ANN as a feature extraction tool rather than as a quantification tool. Also, given the use of PCA is an implicit step of PCR when building the models, it was not used in this case; likewise, for the similarity that would be between PCR and PLS models if it was used, in the case of PLS, it was also discarded in the latter.

After building all the models using the different obtained coefficients as inputs and training them with the training subset, the models' accuracies were evaluated using the samples of the external test subset; the results obtained were also used to carry out the selection of the optimal model in each case (i.e. ANN architecture, number of PCs or latent variables). Once the modelling was completed, its Root Mean Square Error (RMSE) value was calculated and an overall comparison was performed, as shown in Fig. 3 and Table S2. From this, it can be seen that larger errors are obtained when using the whole raw voltammetric data, with similar results for the rest of the cases. Also, it could be seen that, in general, best results are obtained when using DWT as a preprocessing method for the different modelling tools. Moreover, it should be noted that the good results that were obtained employing the *windowed slicing integral*, which represents the implementation of a new and simple preprocessing method. As stated above, it has the disadvantage that the selection of coefficients obtained is arbitrary, thus requiring some previous knowledge or trials to ensure a good departure point. Despite this, it has been proved to be a good way to reduce signal dimensionality, its simplicity being its main advantage.

In addition to RMSE values, comparison graphs of predicted vs. expected concentration for each phenolic compound were built to check the prediction ability of the obtained models, both for train and test subsets. Then slope values and correlation coefficients of the comparison were taken into account in order to help in selecting the best configuration. Therefore, the optimum model would have a small RMSE,

meaning there are no differences between the values predicted by the model and the real ones. Also the comparison graphs should display a correlation value and slope close to 1 and an intercept close to 0, discarding any model error.

One example of these graphs is shown in Fig. 4. It may be seen that a satisfactory trend is obtained in all the cases, with regression lines almost indistinguishable from the theoretical ones. Also, as usual in multivariate calibration models, lower dispersion and uncertainties are obtained for the training subsets.

In this manner, comparison graphs were built for all cases; the obtained regression parameters are summarized in Fig. 5, Table S1 and Table S2. As said, it could be seen that better trend is obtained in all the cases for the training subset, where slopes and correlation coefficients are close to 1 and intercepts closer to 0, with small differences between the various preprocessing and modelling methods. However, despite the low and similar RMSE values obtained between the different modelling methods for the testing subset, higher differences are attained with the comparison graphs. Firstly, it could be seen that sinapic acid is the one presenting the worst models' results in all the cases, implying that its signal is masked by the other compounds.

Leaving that aside, in all cases it could be seen that correlation is still good (r>0.875 in the worst case), but the main difference can be found in the slopes and intercepts (values down to ca. 0.65 and up to 36.5 μM respectively), where a clear decline is obtained particularly with linear models. In this sense, ANNs are the ones presenting the best results with slopes higher than 0.92 and intercepts under 9.5 μM in the worst situation (PCA-ANN).

Hence, for the selection of the best model, both RMSE and comparison graph results must be considered. In this case, the best results were obtained employing FFT-ANN. While in the case of linear models, best results were obtained employing DWT preprocessing of the signal. As stated, these improvements are due to both methods providing a good compression of the data, preserving relevant information and denoising the original signal.

Finally, the good results obtained using *windowed slicing integral method* should be noted, which despite its simplicity proved to be a satisfactory variant. Also, the good performance obtained using CI analysis; although this method has the drawback that the reduced sets are not unique and the arrival to an optimal set depends on the architecture of the ANN used. Moreover, the amount of time required for the feature extraction process is much larger. In spite of this, it allows for the identification of the most relevant sensors and inputs from the voltammograms that contribute to the discrimination and the quantification of the compounds under scrutiny. And consequently allowing the selection of which sensors must be kept for further experiments or the modification of the voltammetric parameters.

Conclusions

In this article, an overview of different preprocessing and processing strategies that can be used when dealing with voltammetric Electronic Tongue data has been presented. In this line, obtained responses were first preprocessed in order to reduce the huge dimensionality of the obtained matrix responses, which allowed for the acquisition of advantages in the modelling stage such as the reduction in training time, avoiding redundancy in input data and obtaining a model with better generalization ability. Afterwards, obtained coefficients were used as inputs in the modelling stage in order to predict the concentration of the desired compounds. Finally, in order to assess the predictive capabilities, RMSE values were calculated and comparison graphs were built, which allowed the selection of the best model.

In order to illustrate the differences between the various methods, the resolution of phenolic antioxidant mixtures was selected as the study case. Numerical results merely present some general trends, given the optimal method could change depending on the study case. However, we want to emphasize that the objective of this article has not been to determine the best technique or set of techniques, but to give an overview of them and to provide the information that would help in the selection of the most suitable one in each case.

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